

Project README

Project Title

Mario-Inspired SDL Game

Project Description

This is a 2D Mario-inspired game developed using SDL (Simple DirectMedia Layer). The player controls a character to collect coins, avoid enemies, and achieve the highest score possible. The game features:

- A jumping mechanic.
- Flying and ground-based enemies.
- Collectible coins with varying values.
- A background soundtrack.

The game ends when the player loses all lives or collides with an enemy.

Requirements

To run this project, you need the following:

1. **SDL Library:** Ensure SDL2 is installed on your system (both video and audio modules).
 - On Linux: Install SDL2 via your package manager (e.g., `sudo apt install libsdl2-dev`).
 - On Windows: Download SDL2 from [SDL's official website](#).
 - The version used during development was SDL2-devel-2.24.0-VC.zip
 - On macOS: Use Homebrew (`brew install sdl2`).
2. **Audio and Image Files:**
 - The following files must be in the same directory as the executable:
 - B24PH1004_B24MT1046_B24PH1008_B24CI1041_B24PH1020_Background.wav (background music).
 - B24PH1004_B24MT1046_B24PH1008_B24CI1041_B24PH1020_Background_Image.bmp (background image).
3. **Score File:**
 - A text file named B24PH1004_B24MT1046_B24PH1008_B24CI1041_B24PH1020_Score.txt must be present in the same folder to store high scores.

How to Run

Follow these steps to compile and run the project:

1. Clone or Download the Project

Download or clone this project to your local machine.

2. Compile the Code

Use a C compiler like GCC to compile the code. Ensure you link SDL2 libraries during compilation.

Example command for compilation:

```
gcc  
B24PH1004_B24MT1046_B24PH1008_B24CI1041_B24PH1020_Mario_Game.c  
-o mario_game -lSDL2 -lSDL2main
```

3. Run the Executable

Execute the compiled file:

```
./mario_game
```

4. Gameplay Instructions

- Use **A** to move left.
- Use **D** to move right.
- Use **W** to jump.
- Avoid enemies and collect coins to increase your score.

Important Notes

1. Audio Playback:
 - Ensure that audio is working on your device. If audio does not replay automatically, re-run the game or check your SDL installation.
2. Game Files Location:
 - All required files (audio, image, and score file) must be in the same folder as the executable for proper functioning.
3. High Scores:

- The game maintains a top 3 high scores list, which is stored in B24PH1004_B24MT1046_B24PH1008_B24CI1041_B24PH1020_Score.txt.
4. Dependencies:
- Ensure SDL2 libraries are properly installed and linked during compilation.

Credits

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